Lecture 12: The Unit Cell and the Neutron Life Cycle

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

February 9, 2026

Reading Assignment

Lamarsh & Baratta (3rd or 4th Edition):

- **Section 6.5:** The Four-Factor Formula (The "Engine" of the reactor).
- **Section 6.8:** Heterogeneous Reactors (Why we lump fuel into rods).

---

1 Introduction

We have discussed cross-sections (σ) for individual nuclei. Now we must build a machine. Real reactors are not homogeneous mixtures of atoms; they are repeating geometric structures called **Lattices**.

Today we define the "Unit Cell"—the smallest repeating volume of fuel and moderator—and trace the life cycle of a neutron within it. This leads to the fundamental figure of merit for the lattice: the **Infinite Multiplication Factor (k_∞)**.

2 The Neutron Life Cycle (The Four Factor Formula)

To model a reactor, we must convert physical processes (fission, resonance capture, leakage) into probabilities.

Imagine an infinite array of unit cells. We track a generation of n thermal neutrons that have just been absorbed in the fuel. We follow them through their "life" until they create the next generation of thermal neutrons.

2.1 1. Fission Production (η)

The cycle begins when thermal neutrons are absorbed by the fuel. However, not every neutron absorbed by the fuel causes fission; some are merely captured (n, γ). We define the **Reproduction Factor η (Eta)** as the number of fission neutrons produced per *absorption* in the fuel:

$$\eta = \nu \frac{\sigma_f}{\sigma_a} = \nu \frac{\sigma_f}{\sigma_f + \sigma_\gamma} \tag{1}$$

- For pure ^{235}U, $\eta \approx 2.08$.
- For natural uranium (mostly ^{238}U, which absorbs but doesn't fission), $\eta \approx 1.3$. This drop is why we generally require enrichment.

$$n \xrightarrow{\text{Fuel Absorption}} n\eta \quad \text{(Fast Neutrons Born)}$$

2.2 2. Fast Fission (ϵ)

The neutrons are born at high energy ($\approx$ 2 MeV). Before they leave the fuel rod, they have a small chance of striking a ^{238}U nucleus and causing a "fast fission" event (as discussed in previous lectures).

- This creates a small "bonus" to our neutron population.
- We define the **Fast Fission Factor ϵ (Epsilon)** as the ratio of total fission neutrons to those produced by thermal fission alone.
- Typically, $\epsilon \approx 1.03$ (a 3% bonus).

$$n\eta \xrightarrow{\text{Fast Fission}} n\eta\epsilon$$

2.3 3. Slowing Down & Resonance Escape (p)

The neutrons leave the rod and enter the moderator (water). They must slow down from 2 MeV to 0.025 eV.

- **The Danger Zone:** As they pass through intermediate energies (1–1000 eV), they face the "Valley of Death"—the giant resonance peaks of ^{238}U. If they touch fuel at this energy, they are captured without fission.
- We define the **Resonance Escape Probability p** as the probability that a neutron successfully reaches thermal energy.
- In a typical LWR, $p \approx 0.7$ (meaning 30% of neutrons are lost here).

$$n\eta\epsilon \xrightarrow{\text{Slowing Down}} n\eta\epsilon p \quad \text{(Thermal Neutrons Reached)}$$

2.4 4. Thermal Utilization (f)

Now the neutrons are thermal. They diffuse through the lattice. They can be absorbed by the fuel (useful) or by the water/steel/control rods (waste). We define the **Thermal Utilization Factor f** as:

$$f = \frac{\Sigma_{a,fuel}}{\Sigma_{a,total}} = \frac{\text{Neutrons Absorbed in Fuel}}{\text{Total Neutrons Absorbed}} \tag{2}$$

$$n\eta\epsilon p \xrightarrow{\text{Diffusion}} n\eta\epsilon p f \quad \text{(New Generation)}$$

2.5 The Result: k_∞

The ratio of the new generation (n') to the old generation (n) is the infinite multiplication factor:

$$k_\infty = \frac{n'}{n} = \epsilon p f \eta \tag{3}$$

3 The "Rod" Problem: Why Heterogeneous?

Why do we construct reactors with fuel rods? Why not just dissolve uranium salts in water (Homogeneous)?

The Physics of Lumping:

1. **Resonance Protection (Improving p):** ^{238}U is incredibly "hungry" for neutrons at resonance energies. If you mix U and H atoms uniformly, the neutrons never escape. By lumping the fuel into rods, we force the neutrons to slow down in the *water*, physically separated from the ^{238}U.

2. **Thermal Disadvantage (Hurting f):** Lumping fuel makes it harder for *thermal* neutrons to get back in. The outer skin of the rod "shields" the inside.

4 Design Insight: How Big Should the Rod Be?

A student asked: *"What is the optimal radius for a fuel rod?"* This is a "Goldilocks" problem determined by the Mean Free Path of the neutron.

Quick Recall: The Mean Free Path

Before we optimize the rod size, recall from our discussion on cross-sections that the average distance a neutron travels before colliding with a nucleus is:

$$\lambda = \frac{1}{\Sigma_t} \tag{4}$$

where Σ_t is the macroscopic total cross-section.

- **In Fuel (Fast Neutrons):** Σ_t is small ($\lambda \approx 6$ cm).
- **In Fuel (Thermal Neutrons):** Σ_t is large ($\lambda \approx 1.5$ cm).

The Optimization Constraints

- **Constraint 1: Fast Neutron Escape ($\lambda_{fast} \approx 6$ cm)** Fission neutrons are born fast (2 MeV). They need to escape the rod to reach the water. Since $\lambda_{fast} \approx 6$ cm, a standard 1 cm rod is transparent to them. They fly out easily.

- **Constraint 2: Thermal Neutron Penetration ($\lambda_{thermal} \approx 1 - 2$ cm)** Thermal neutrons must diffuse back *into* the rod to cause fission. Fuel is very absorbent. If the rod is too thick (e.g., 6 cm), the neutrons will all be eaten on the surface, and the center will generate no power.

- **Constraint 3: The Thermal Limit (Melting)** Uranium Oxide (ceramic) is a poor heat conductor. If the rod is too thick, the heat cannot escape, and the centerline will melt ($T_{melt} \approx 2800°$C).

The Solution: Commercial fuel rods are almost universally $\sim$ **1** cm in diameter.

- Small enough for heat and thermal neutrons to get out/in.
- Large enough to provide structural integrity and fuel volume.

5 Cladding: The Critical Shell

5.1 Why do we need it? (The problem with UO_2)

While Uranium Dioxide (UO_2) is the standard fuel due to its high melting point (2865°C) and radiation stability, mechanically and chemically, it is terrible.

- **It is a Ceramic:** Like a dinner plate, it is brittle. Under thermal stress, the fuel pellets crack and fragment.
- **Fission Products:** As fission occurs, the crystal lattice fills with foreign contaminants. Gaseous products (Krypton and Xenon) form bubbles, causing the pellet to swell and potentially crumble.
- **Solubility:** If the hot ceramic comes into direct contact with the coolant (water), it can erode, releasing highly radioactive fission products directly into the turbine loop.

To solve this, we must hermetically seal the fuel pellets in a metal tube known as **Cladding**.

5.2 The Material Search: A "Goldilocks" Problem

Selecting a material for fuel cladding requires balancing two competing requirements:

1. **Nuclear Transparency:** The material must have a low thermal neutron capture cross-section (σ_a) to avoid "poisoning" the chain reaction.
2. **Thermal Resilience:** The material must maintain structural integrity at high operating temperatures (> 300°C) and withstand potential accident conditions.

Table 1: Thermal neutron cross-sections and melting points of potential cladding metals.

Metal	**Absorb.** σ_a **[barns]**	**Melt Point** [°C]	**Verdict / Limitation**
Beryllium (Be)	0.009	1287	Too Brittle & Toxic
Magnesium (Mg)	0.063	650	Corrodes in Water / Low T_m
Zirconium (Zr)	**0.185**	**1855**	**Optimal Balance**
Aluminum (Al)	0.230	660	Melts too easily (Low T_m)
Iron (Fe)	2.560	1538	Parasitic Neutron Absorption
Titanium (Ti)	6.100	1668	Excessive Neutron Absorption

5.3 Analysis of Candidates

The "Winners" that Lost (Be, Mg, Al): Looking at the table, Beryllium and Magnesium are superior to Zirconium in terms of neutron economy. However, Beryllium is mechanically unsuitable (brittle) and hazardous. Magnesium (used in early gas-cooled reactors) and Aluminum cannot withstand the high temperatures and corrosive environment of modern water-cooled reactors; their melting points are dangerously close to accident temperatures.

The Iron Problem: Iron (steel) is structurally excellent but acts as a "neutron hog." With a cross-section of 2.56 barns (over $10\times$ that of Zirconium), using steel cladding would require significantly higher uranium enrichment to compensate for the lost neutrons, making the fuel cycle uneconomical.

Why Zirconium Wins: Zirconium is the only candidate that provides "good enough" neutron transparency (≈ 0.185 b) while possessing a melting point high enough (1855°C) to serve as a safety barrier.

5.4 The Hafnium Contamination Issue

There is one major catch. In nature, Zirconium is always found combined with **Hafnium** (Hf). They are "chemical twins" (located in the same column of the periodic table) and are extremely difficult to separate.

- **Zirconium:** $\sigma_a = 0.185$ barns (Transparent).
- **Hafnium:** $\sigma_a \approx 104$ barns (Black).

Natural Zirconium typically contains 1–3% Hafnium. If you built a reactor out of "Commercial Grade" Zirconium, the Hafnium impurity would poison the reaction, making criticality impossible.

This leads to the distinction between two grades of metal:

- **Commercial Grade Zr:** Contains Hf. Used in chemical plants. Cheap.
- **Nuclear Grade Zr:** Hf removed via difficult liquid-liquid extraction. Expensive, but required for cladding.

Ironically, the separated Hafnium is so absorbent that we use it to make the Control Rods!

Historical Note: The Zirconium-Hafnium Engineering Choice

The use of Zirconium in nuclear reactors was not obvious; it was forced into existence by Admiral Hyman G. Rickover for the *USS Nautilus*.

In the late 1940s, published data suggested Zirconium had a high neutron cross-section, making it unsuitable for reactors. Researchers at MIT and Oak Ridge National Laboratory (ORNL) suspected this data was flawed and discovered the culprit: the inevitable 2% Hafnium impurity found in all natural Zirconium ore.

When Rickover was told that separating the two "chemical twins" was commercially impossible, he refused to accept the answer. He drove the development of a liquid-liquid extraction process (using methyl isobutyl ketone) at the Y-12 plant in Oak Ridge. He then forced the Bureau of Mines and the Wah Chang Corporation to scale the process from grams to tons. This effort dropped the price of reactor-grade Zirconium from over \$300/lb to under \$5/lb, making the nuclear navy possible.

Reference: Rickover, H. G. (1975).*History of the development of zirconium alloys for use in nuclear reactors*. U.S. Energy Research and Development Administration, Division of Naval Reactors. This reference concludes with the famous Rickover engineering mantra: *"The Devil is in the details, but so is salvation".*

Technical Note: Reactor Decay Chains

Nuclear reactor physics is governed by four distinct categories of transmutation chains:

1. **Breeding Chains:** The conversion of fertile material into fissile fuel.
2. **Actinide Loss Chains:** The parasitic capture reactions that degrade fuel quality.
3. **Fission Product Chains:** The decay of split fragments into poisons or long-lived waste.
4. **Activation Chains:** The irradiation of non-fuel materials (coolant, moderator, structure).

1 1. The Breeding Chains (Fertile $\rightarrow$ Fissile)

These chains allow non-fissile isotopes to absorb a neutron and eventually decay into fissile fuel.

The Uranium-Plutonium Cycle (Standard LWR/Fast Breeder)

The primary mechanism for plutonium production in commercial reactors.

$$\begin{aligned} {}^{238}_{92}\mathrm{U} + n \longrightarrow {}^{239}_{92}\mathrm{U} &\xrightarrow[\mathbf{23.5}\text{ min}]{\beta^-} {}^{239}_{93}\mathrm{Np} \\ &\xrightarrow[\mathbf{2.36}\text{ days}]{\beta^-} {}^{239}_{94}\mathrm{Pu} \quad (\textbf{Fissile}) \end{aligned}$$

The Thorium-Uranium Cycle

An alternative fuel cycle utilizing the abundant Thorium-232. Note the long half-life of the intermediate Protactinium, which poses a "storage" challenge during operation.

$$\begin{aligned} {}^{232}_{90}\mathrm{Th} + n \longrightarrow {}^{233}_{90}\mathrm{Th} &\xrightarrow[\mathbf{22.3}\text{ min}]{\beta^-} {}^{233}_{91}\mathrm{Pa} \\ &\xrightarrow[\mathbf{27.0}\text{ days}]{\beta^-} {}^{233}_{92}\mathrm{U} \quad (\textbf{Fissile}) \end{aligned}$$

2 2. The Actinide Loss Chains ("Parasitic" Reactions)

Not every absorption results in fission or useful breeding. Some reactions create "dead ends" that remove neutrons from the economy and pollute the fuel with non-fissile isotopes.

Uranium-236 Buildup (The Dead End)

About 15% of thermal neutrons absorbed by U-235 result in capture rather than fission ($\alpha = \sigma_\gamma/\sigma_f \approx 0.17$).

$${}^{235}_{92}\mathrm{U} + n \longrightarrow {}^{236}_{92}\mathrm{U} \quad (\text{Long-lived Waste, } T_{1/2} \approx 23\text{ My})$$

Impact: U-236 is a neutron absorber. Recycled uranium must be enriched higher to compensate for this "poison."

The "Dirty" Plutonium Chain

If Fissile Pu-239 sits in the reactor too long, it captures another neutron to become Pu-240.

$$^{239}_{94}\text{Pu} + n \longrightarrow {}^{240}_{94}\text{Pu} \quad \text{(Spontaneous Fission Source)}$$

Impact: Pu-240 is not fissile by thermal neutrons and has a high spontaneous fission rate, making the plutonium unsuitable for weapons (the definition of "Reactor Grade").

The Americium Pathway

If Pu-240 captures *another* neutron, it becomes fissile Pu-241. However, Pu-241 decays relatively quickly into Americium.

$$^{241}_{94}\text{Pu} \xrightarrow[\mathbf{14.3}\text{ yr}]{\beta^-} {}^{241}_{95}\text{Am} \quad \text{(Strong Neutron Absorber)}$$

Impact: Am-241 is a major contributor to long-term radiotoxicity in nuclear waste.

3 3. Fission Product Chains (Poisons & Waste)

Fission fragments decay toward the valley of stability. Some intermediates have massive cross-sections (Poisons).

The Xenon-135 Chain (Transient Poison)

The dominant short-term operational constraint.

$$\text{Fission} \longrightarrow {}^{135}_{52}\text{Te} \xrightarrow{19\text{ s}} {}^{135}_{53}\text{I} \xrightarrow[\mathbf{6.6}\text{ hr}]{\beta^-} {}^{135}_{54}\text{Xe} \xrightarrow[\mathbf{9.2}\text{ hr}]{\beta^-} {}^{135}_{55}\text{Cs}$$

- σ_a(Xe-135) $\approx 2.6 \times 10^6$ barns.
- Causes the "Iodine Pit" after shutdown.

The Samarium-149 Chain (Stable Poison)

A permanent poison that accumulates until equilibrium.

$$\text{Fission} \longrightarrow {}^{149}_{60}\text{Nd} \xrightarrow{1.7\text{ hr}} {}^{149}_{61}\text{Pm} \xrightarrow[\mathbf{53}\text{ hr}]{\beta^-} {}^{149}_{62}\text{Sm} \quad (\textbf{Stable})$$

4 4. Coolant & Structural Activation

Neutrons leak from the fuel and activate the environment.

Nitrogen-16 (Water Activation)

The primary source of radiation in the coolant loop during operation.

$$^{16}_{8}\text{O} + n \longrightarrow {}^{16}_{7}\text{N} + p \quad \text{(Fast Neutron Reaction)}$$

- **Decay:** $^{16}\text{N} \xrightarrow{7.1\text{ s}} {}^{16}\text{O} + \gamma$ (6 MeV).
- Creates high-energy gamma field, but vanishes seconds after shutdown.

Sodium-24 (Fast Reactor Coolant)

Sodium cooled reactors must deal with this activation product.

$$^{23}_{11}\text{Na} + n \longrightarrow {}^{24}_{11}\text{Na} \xrightarrow[\mathbf{15}\text{ hr}]{\beta^-} {}^{24}_{12}\text{Mg} + 2\gamma$$

Impact: Unlike water (N-16), Sodium-24 remains radioactive for days, complicating maintenance.

Tritium Production (^{3}H)

Tritium is a mobile, radioactive isotope of hydrogen, produced via:

1. **Boron Shim Control:** $^{10}\text{B} + n \longrightarrow 2\alpha + {}^3\text{H}$ (Rare branch) or $^{10}\text{B}(n,\alpha)^7\text{Li} \xrightarrow{n} {}^3\text{H}$.
2. **Deuterium Capture:** $^2\text{H} + n \longrightarrow {}^3\text{H}$ (Heavy water reactors).
3. **Lithium Contaminants:** $^6\text{Li} + n \longrightarrow {}^4\text{He} + {}^3\text{H}$.